

C Programming

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C Programming

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Q1. What is the purpose of the `main()` function in a C program? Explain its significance.

The `main()` function is the entry point of a C program. Execution always starts from `main()`. It controls the flow of the program and usually returns an integer value (commonly 0) to indicate successful execution.

Q2. Explain the difference between a variable declaration and a variable initialization in C.

- Declaration: It tells the compiler the type and name of a variable, but does not assign a value.

Example: `int a;`

- Initialization: It assigns a value to a variable at the time of declaration.

Example: `int a = 10;`

Declaration creates the variable, while initialization gives it an initial value.

Q3. Write a C program to display a personalized greeting message. (Should contain 'hello' or 'welcome' in the message)

```
#include <stdio.h>

int main(){

    printf("hello charlie\n");

    return 0;

}

PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src> cd
; if ($?) { gcc q3.c -o q3 } ; if ($?) { .\q3 }
hello charlie
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>
```

Q4. What are the different data types available in C? Provide examples of each data type.

C provides different data types to store different kinds of values:

- int – Stores whole numbers (e.g., int age = 18;)
- float – Stores decimal numbers (e.g., float price = 9.5;)
- double – Stores larger or more precise decimal numbers (e.g., double pi = 3.141592;)
- char – Stores a single character (e.g., char grade = 'A';)
- void – Represents no value (used in functions)

These data types help manage memory efficiently.

Q5. Explain the concept of type conversions in C. Provide examples of implicit and explicit type conversions.

Type conversion is the process of converting one data type into another.

- Implicit conversion: Done automatically by the compiler when mixing different data types.
Example: Converting int to float automatically.
- Explicit conversion (Type Casting): Done manually by the programmer using casting.
Example: (float)a

Type conversion is important to avoid data loss and get correct results in calculations.

Q6. Write a C program to calculate the area of a rectangle. Prompt the user to enter the length and width, and display the result.

```
int main() {  
  
    int l,b,a;  
  
    printf("Enter the Length of the rectangle: ");  
    scanf("%d", &l);  
  
    printf("Enter the Width of the rectangle: ");  
    scanf("%d", &b);
```

```

a = l*b;

printf("The area of the rectangle is: %d\n", a);

return 0;
}
tempCodeRunnerFile.c:6:5: note: include '<stdio.h>' or provide a declarati
Enter the Length of the rectangle: 4
Enter the Width of the rectangle: 6
The area of the rectangle is: 24

```

Q7. What is the role of the scanf() function in C? Provide an example of its usage.

The scanf() function in C is used to take input from the user through the keyboard. It reads the input data and stores it in the variables provided. It uses format specifiers like %d for integers, %f for floats, and %c for characters. The & symbol is used before the variable name to store the input at its memory address.

Example:

```
#include <stdio.h>
```

```
int main() {
```

```
    int age;
```

```
    printf("Enter your age: ");
```

```
    scanf("%d", &age);
```

```

    printf("You entered: %d", age);

    return 0;

}

```

In this example, scanf() reads an integer entered by the user and stores it in the variable age.

Q8. Write a C program to convert temperature from Celsius to Fahrenheit. Prompt the user for a temperature in Celsius and display the equivalent temperature in Fahrenheit. (Formula:

$$\text{fahrenheit} = (\text{celsius} * 9 / 5) + 32)$$

```

#include <stdio.h>

int main()
{
    int c, f;

    printf("Enter the value of temepreature in celcius:");

    scanf("%d",&c);

    f=(c*9/5)+32;

    printf("The value of temepreature in Farahrenheit is: %d\n", f);

    return 0;

}

; if ($?) { gcc tempCodeRunnerFile.c -o tempCodeRunnerFile } ; if ($?) { .\
Enter the value of temepreature in celcius:100
The value of temepreature in Farahrenheit is: 212
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-camitkc750-del\src>

```

Q9. Input a number representing days and print the equivalent number of weeks and days.

(Example: 10 days = 1 week and 3 days)

```

#include <stdio.h>

#include <conio.h>

int main(){

    int days, weeks, remaining_days;

    printf("Enter the number of days: ");

    scanf("%d", &days);

    weeks = days / 7; // Calculate the number of weeks

    remaining_days = days % 7; // Calculate the remaining days

    printf("%d week(s) and %d day(s)\n", weeks, remaining_days);

return 0;

}

```

```

; if ($?) { gcc q9.c -o q9 } ; if ($?) { .\q9 }

```

```

Enter the number of days: 45

```

```

6 week(s) and 3 day(s)

```

Q10. Write a C program to swap the values of two variables using a temporary variable.

```

#include <stdio.h>

int main() {

    int a, b, temp;

    printf("Enter the value of a and b: ");

    scanf("%d %d", &a, &b);

    temp = a;

```

```

a = b;

b = temp;


printf("After swapping, the value of a is: %d\n", a);

printf("After swapping, the value of b is: %d\n", b);


return 0;
}

```

Enter the value of a and b: 4 5
 After swapping, the value of a is: 5
 After swapping, the value of b is: 4
 PS C:\Users\Lenovo\OneDrive\Desktop> c:\assignment_1_ramitkc750_del\src>

Q11. Write a C expression that performs the following operations in a single line: increment a variable by 1, multiply it by 3, and subtract 10.

```

#include <stdio.h>

int main()
{
    int a,x;

    printf ("enter the value of a :");

    scanf("%d",&a);

    x=((a+1)*3)-10;

    printf("the value of x is %d",x);

    return 0;
}

```



```

enter the value of a :7
the value of x is 14
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

```

Q12. Given three variables a, b, and c, write an expression that checks if a is greater than b and c is not equal to 0.

```

#include <stdio.h>

int main(){
    int a,b,c;

    printf("Enter the value of a, b and c:");

    scanf("%d,%d,%d", &a, &b, &c);

    if(a>b && c!=0)

        printf("The expression is true \n");

    else

        printf("The expression is false \n");

    return 0;
}

```

```

Enter the value of a, b and c:5 6 7
The expression is true
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

```

Q13. Write a C expression that evaluates whether a number is divisible by both 2 and 3 (without using the modulus operator).

```

#include <stdio.h>

int main()

```

```

{
    int num;

    printf("Enter a number:");

    scanf("%d",&num);

    if(num/2*2==num && num/3*3==num)

        printf("The number is divisible by both numbers");

    else

        printf("The number is not divisible by both numbers");

return 0;
}

```

Enter a number: 6
The number is divisible by both numbers

PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

Q14. Create an expression that swaps the values of two variables x and y without using a temporary variable.

```

#include<stdio.h>

int main()
{
    int x, y;

    printf("Enter the value of x and y:");

    scanf("%d%d", &x, &y);

    x=x+y;

    y=x-y;

    x=x-y;
}

```

```

printf("After swapping, value of x is: %d\n", x);

printf("After swapping, value of y is: %d\n", y);

return 0;
}

```

Enter the value of x and y: 5 6
 After swapping, value of x is: 6
 After swapping, value of y is: 5
 PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

Q15. Write an expression that checks if a number is both positive and even.

```

#include<stdio.h>

int main()
{
    int a;

    printf("Enter the value of a:");

    scanf("%d",&a);

    if(a>0 && a%2==0)

        printf("The number is both positive and even \n");

    else

        printf("The number is not both positive and negative \n");

    return 0;
}

```

Enter the value of a: 6
 The number is both positive and even
 PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

Q16. Given two variables x and y, write an expression that calculates the average of their values.

```
#include <stdio.h>

int main()
{
    float x, y, average;

    printf("enter the value of x and y:");

    scanf("%f%f", &x, &y);

    average=(x+y)/2;

    printf("The average of x and y is %f\n", average);

    return 0;
}

; IT ($?) { gcc q16.c -o q16 } ; IT ($?) { ./q16 }
enter the value of x and y:6 7
The average of x and y is 6.500000
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment_1_ramitkc750-del\spr>
```

Q17. Create an expression that checks if a given character is an uppercase letter.

```
#include <stdio.h>

int main(){
    char a;

    printf("Enter a character");

    scanf("%c", &a);

    if(a>='A' && a<='Z')

        printf("The character is an uppercase letter: %c\n", a);

    else
```

```
printf("The character is a lowercaseletter: %c\n", a);

return 0;

}
```

Enter a charactera\
The character is a lowercaseletter: a
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

Q18. Write a C expression that calculates the sum of the squares of three different numbers.

```
#include<stdio.h>

int main(){

    int a,b,c,sum;

    printf("Enter the value of a, b and c:");

    scanf("%d%d%d", &a, &b, &c);

    sum=a*a+b*b+c*c;

    printf("The sum of squares is: %d\n", sum);

    return 0;

}
```

Enter the value of a, b and c:4 5 6
The sum of squares is: 77
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

Q19. Given three variables a, b, and c, write an expression that checks if a is equal to b and b is not equal to c.

```
#include <stdio.h>

int main(){

    int a,b,c;
```

```

printf("Enter the value of a, b and c:");

scanf("%d%d%d", &a, &b, &c);

if(a==b && b!=c)

printf("Expression is true \n");

else

printf("Expression is false \n");

return 0;

}

```

Enter the value of a, b and c:4 5 6
Expression is false

Q20. Write an expression that checks if a number is a multiple of either 3 or 5.

```

#include <stdio.h>

int main()
{
    int a;

    printf("Enter the value of a:");

    scanf("%d",&a);

    if(a%3==0 || a%5==0)

        printf("The number is a multiple of either one");

    else

        printf("The number is not a multiple of either one");
}

```

```

    return 0;
}

```

Enter the value of a:8
The number is not a multiple of either one
PS C:\Users\Lenovo\OneDrive\Desktop> cd assignment_1_pawitkc750_dcl\src

Q21. Create an expression that swaps the values of three variables x, y, and z in a cyclic order (i.e., x becomes y, y becomes z, and z becomes x).

```

#include<stdio.h>

int main(){
    int x,y,z,temp;

    printf("Enter the value of x, y and z:");

    scanf("%d%d%d", &x, &y, &z);

    temp = x;
    x = y;
    y = z;
    z = temp;

    printf("After swapping, the value of x is: %d\n", x);
    printf("After swapping, the value of y is: %d\n", y);
    printf("After swapping, the value of z is: %d\n", z);

    return 0;
}

```

```
Enter the value of x, y and z:3 4 5
After swapping, the value of x is: 4
After swapping, the value of y is: 5
After swapping, the value of z is: 3
```

PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src> █

Q22. Write a C expression that calculates the square root of the sum of two numbers, rounded to the nearest integer.

```
#include <stdio.h>

#include<math.h>

int main(){

    double a = 9.0, b = 16.0;

    double sum = a + b;

    double result = sqrt(sum);

    int rounded = (int)(result + 0.5); // rounding to nearest integer

    printf("The square root of the sum is: %d\n", rounded);

    return 0;

}
```

```
5 11 (p:) [ 800 444.0 0 444 ] 5 11 (p:) [ 0.444 ]
```

```
Enter the value of a and b:6 7
the square root of the sum is: 3
```

PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src> █

Q23. Given a variable num, write an expression that checks if it is a power of 2.

```
#include <stdio.h>

#include<math.h>

int main()
```



```

{
    int a;

    printf("Enter the value of a:");

    scanf("%d", &a);

    if(a>0 && (a &(a-1) )==0)

        printf("The number is a power of 2:%d\n",a);

    else

        printf("The number is not a power of 2:%d\n",a);

    return 0;
}

```

Enter the value of a:7
The number is not a power of 2:7
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src>

Q24. Create an expression that checks if a given number is a perfect square.

```

#include <stdio.h>

#include<math.h>

int main(){

    int num = 16;

    int sqrt_num = (int)sqrt(num);

    if(sqrt_num * sqrt_num == num)

        printf("The number is a perfect square: %d\n", num);

    else

        printf("The number is not a perfect square: %d\n", num);
}

```

```
return 0;
```

```
}
```

```
Enter the value of s:78
```

```
The number is not a perfect square:78
```

```
PS C:\Users\Lenovo\OneDrive\Desktop\c\assignment-1-ramitkc750-del\src> |
```